

Week 1_Practice 4_Vocabulary

Task 1. Complete each sentence using the most appropriate expression from the box. Two expressions are extra.

1 2 3 4 5
anticipatory competence • cutting-edge digital skills • emerge from the data • facilitate • gain
consensus • ill-defined • keep pace with progress • transcend disciplinary boundaries •
transversal skills • validate the initial findings

1. When developing AI systems for sustainable cities, IT specialists need anticipatory to consider not only current requirements but also possible long-term technological, environmental and social consequences.
2. The relationship between digital inclusion and access to public services did not become immediately apparent; it began to 3 only after researchers had analysed the complete dataset.
3. Addressing climate change through digital innovation often requires engineers to 8 and collaborate with environmental scientists, economists and policymakers.
4. Technical expertise alone may be insufficient. Communication, critical thinking and teamwork are important 9 that can be applied across professional contexts.
5. Because the original problem statement was too 6, the development team could not determine which sustainability outcomes the proposed platform was expected to achieve.
6. Organisations must continuously update employees' digital competence if they are to 7 in rapidly changing areas such as AI, cybersecurity and data analytics.
7. Before recommending nationwide implementation of the digital platform, researchers conducted a second study to 10.
8. Effective data visualisation can 4 communication between technical specialists, policymakers and community representatives.

Task 2. Choose A, B, C or D. Pay attention to meaning and academic collocation.

1. A research team wants to determine whether the results of a smart city pilot in Astana could also apply to other cities. The researchers need to:

- A. replicate technological progress
- ☒ B. assess the transferability of the findings
- C. envision academic rigour
- D. gain transversal skills

2. An IT researcher presents empirical evidence to argue that digital accessibility should become a central requirement in public-sector software. The researcher is:

- ☒ A. making the case for digital accessibility
- B. keeping pace with digital accessibility
- ~~C. embedding the findings~~
- D. replicating the importance of accessibility

3. A project leader succeeds in securing broad agreement among three stakeholder groups after extensive discussion. The project leader has:

- A. transcended the findings
- B. facilitated a disagreement

☒ C. gained consensus

D. validated a boundary

4. Researchers repeat an earlier study with a new group of software-engineering students to determine whether similar results can be obtained. They:

☒ A. replicate the study

B. envision the study

C. embed the study

D. perceive the study as critical

5. A university integrates sustainability competences directly into its software-engineering curriculum rather than teaching them as a separate optional topic. It:

A. emerges sustainability competences

☒ B. embeds sustainability competences in the curriculum

C. transcends the curriculum

D. attests the curriculum

6. Employers argue that knowledge of AI ethics and sustainable computing is essential rather than merely desirable. They:

☒ A. perceive these competences as critical

B. perceive these competences critically

C. facilitate these competences as critical

D. replicate these competences

7. A study uses transparent methodology, systematic analysis and carefully justified conclusions. It can be said to:

A. have technological convergence

☒ B. demonstrate academic rigour

C. gain academic consensus

D. facilitate academic evidence

Task 3. Complete each sentence with one word from the box to form the correct academic expression. Use each word once.

boundaries • consensus • critical • findings • importance • issues • progress • rigour

1. The research team repeated the analysis using an independent dataset to validate the initial

4.

2. Software engineers must keep pace with technological 7 if they are to work effectively with rapidly evolving AI systems.

3. The study demonstrates academic 8 through transparent procedures, systematic analysis and clearly justified conclusions.

4. The project required computer scientists to transcend disciplinary 1 and work with environmental researchers and policymakers.

5. After several rounds of consultation, the project leader was able to gain 2 among the participating stakeholder groups.

6. The evidence appears to attest to the 5 of accessible design in digital public services.

7. Before deployment, the development team must address pertinent 6 such as privacy, accessibility and environmental impact.

8. Employers increasingly perceive sustainability-related digital competences as 3 for future IT professionals.

Task 4. Professional IT Scenarios — Select the Best Expression

Read each scenario and choose the expression that best completes it. Two expressions are not needed.

a certain degree of ¹convergence • ²capacity building • ³cutting-edge digital skills • ⁴evidence-based policy • ⁵equity and inclusion • ⁶facilitate • ⁷fall behind technological developments • ⁸ill-defined • stakeholder engagement • sustainable solution

1. A university describes its AI project simply as “using technology to improve education”, without identifying the target users, specific problem, intended outcome or relevant SDG. The project objective is 8.

2. A public organisation introduces advanced digital services but provides no continuing professional development for its IT staff. Over time, employees may 7.

3. A digital-health platform is designed for screen readers, low-bandwidth connections and users with different levels of digital literacy. The design prioritises 5.

4. Software engineers, environmental specialists and urban planners begin with different priorities, but their recommendations become increasingly similar after reviewing the evidence. This indicates 4.

5. Before deploying a generative-AI service, the company consults users, regulators, cybersecurity specialists and accessibility experts. This is an example of 9.

6. A government funds systematic training, mentoring and technical resources so that teachers can use digital tools effectively. This programme represents 2.

7. A ministry decides whether to expand an AI-supported public service only after examining reliable data on accessibility, cost, user outcomes and environmental impact. This is an example of 4.

8. A smart-energy platform produces sustained reductions in energy use while remaining financially viable and accessible to users. It can therefore be described as a 10.